

JOU-CHEN (Anna) LIU

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Education

Master Double Degree Program

Sep. 2023 – May 2026

Karlsruhe University of Applied Science – Germany | German grade: 1.7 (good)

- Master of Science (MSc) in Business Information Systems

National Taiwan University of Science and Technology – Taiwan, R.O.C | GPA: 3.99

- Master of Business Administration (MBA) in Information Management

National Central University – Taiwan, R.O.C | GPA: 3.59

Sep. 2019 – June 2023

- Bachelor of Business Administration (BBA) in Information Management
- Minor Specialty: Programming Design in the Department of Computer Science Information Engineering

Languages & IT Skills

Mandarin: Native / **English:** Fluent, Tests dispensed by ETS: TOEIC 760 | CEFR B1

IT Software: Python, Java, C#, C, C++, HTML, CSS, JavaScript, Vue, React, MySQL, SQL, Microsoft SQL Server, Android Studio, Google Cloud Platform, Firebase, Virtual Machine, Linux

Technical skills: Software Engineering, Software Testing, Full-Stack Development, Data Analysis, Machine Learning, Computer Vision, Large Language Model, Natural Language Processing

Work Experience

ADD Clothes (ADD 服飾)

Piecework Software Engineer

Apr. 2023 – Now

- Inventory and Sales System: Automated Excel data extraction and custom report generation via Python
- Report Email System: Developed Python-based 1-click SMTP/MIME desktop app sending reports to 40 vendors
- Impact: Resolved legacy software failures, achieving stability that attracted peer acquisition interest

Multicast Information Co., Ltd. (銓聯資訊有限公司)

Information Assistant Intern

Apr. 2023 – Aug. 2023

- Full-Stack Web Development, Android Studio APP Development, Data Processing, System Document Writing

Master Thesis

Test Cases Generation with CodeT5+ and Multi-Objective Reinforcement Learning

- Developed an automated Test-Driven Development (TDD) framework integrating CodeT5+ model and Multi-Objective Reinforcement Learning (MORL) to generate test cases directly from natural language requirements
- Optimized test case quality using Proximal Policy Optimization (PPO) and Pareto front analysis based on syntax, executability, and coverage rewards
- Fine-tuned a lightweight CodeT5+ (770M), dropping validation loss (8.76 → 0.45) to validate AI-assisted TDD

College Graduation Project

(APP) CAN - Music Emotion Recognition Application

Feb. 2022 – Nov. 2022

- Built an application cooperating the topics of music and emotion, and facilitating the user experience of services on the online streaming platform
- Applied skills such as fine-tuning BERT model, BeautifulSoup, Librosa, K-means Clustering
- International ICT Innovative Services Awards 2022 – **Excellent Work Award**

Leadership & Volunteer Experience

Curation Team Leader, MIS Graduation Project Carnival, National Central University

Team Captain, MIS Women's Volleyball team, National Central University

Volunteer Team Leader, Kind Kids Group, National Central University

Activities Team Leader, 2023 Tzu Chi Youth Global Humanities Education Exchange, Taiwan & Malaysia